

Iowa Liquor Retail Sales Analysis

SQL Analysis using Google BigQuery

Analysis period	2021 to 2025
Analysis records	12,908,429
Platform	Google BigQuery Standard SQL
Project focus	Sales trends, category mix, products, counties and retailer locations

Executive summary

This project analyzes Iowa liquor retailer purchasing activity from 2021 through 2025 using a curated BigQuery view containing 12.9 million records. The analysis evaluates market trends, seasonality, category and product performance, geographic concentration, and leading retailer locations. Sales peaked at \$447.24 million in 2024 before declining by approximately 5.0% to \$424.99 million in 2025.

The 2025 decline was broad rather than isolated: sales were lower than 2024 in 10 of 12 months. Canadian Whiskies recorded the largest decline among the five leading categories, while 100% Agave Tequila remained the highest-value category on a net-sales-per-bottle basis. Polk County was the largest county market, and the two leading retailer locations were also based there.

The findings support focused recovery actions for weakening categories, differentiated strategies for high-value and high-volume products, careful geographic prioritization, and monitoring of retailer concentration. Recommendations are limited to purchasing and sales evidence because the dataset does not include profitability, inventory, promotions, customer demographics, or population.

1 Project objective and business questions

The objective was to build an SQL analysis that converts a large public retail dataset into a concise set of business findings and recommendations. The project intentionally uses explainable SQL and a limited number of high-value business queries.

- How did annual sales, bottles sold, volume, stores, and product coverage change from 2021 to 2025?
- Was the 2025 sales decline seasonal or broad across the year?
- Which liquor categories and product SKUs generated the highest net sales?
- Which leading categories gained or lost sales, and which generated more value per bottle?
- Which counties and retailer purchasing locations contributed the most sales?

2 Analytical scope and workflow

The source table contained 34,016,839 records dated from 3 January 2012 through 30 April 2026 at the time of profiling. The analysis was restricted to the five complete calendar years from 2021 through 2025. The partial 2026 period was excluded to prevent incomplete-year comparisons.

1. Profile the source table, date coverage, schema, and distinct business entities.
2. Validate completeness, negative values, zero prices, and location fields.
3. Create the reusable view `portfolio-project-508812.iowa_liquor_portfolio.vw_sales_2021_2025`.
4. Validate the view: 12,908,429 records from 4 January 2021 through 31 December 2025.
5. Run eight focused business-analysis queries against the curated view.

3 Data quality assessment and treatment

Quality checks were performed before business analysis. Records were retained unless a specific field made them unsuitable for a particular calculation. This preserves statewide net sales while preventing unsupported replacements or assumptions.

Check	Result	Treatment
Critical transaction fields	0 missing for invoice item, store, item, category, bottles and sales	Retained all records
Vendor number and name	4 missing	Retain; label Unknown only in vendor analysis
Negative bottles and sales	10,144 rows; both values negative together	Retain as likely returns or corrections
State bottle retail	48 zero values; 0 negative values	Keep for sales; exclude from retail-price calculations
Address, city and county	5,138 missing; ZIP missing in 5,160 rows	Keep; exclude unassigned rows from geographic rankings
Store coordinates	564,720 missing	Do not use exact coordinates in core analysis

Interpretation principles

- Negative quantities and sales remain in totals so reported values represent net activity rather than overstated gross sales.
- Large measures are reported in millions for readability, while store and product counts remain whole numbers.
- Annual totals come from unrounded source values. Small differences can occur when displayed monthly values are manually added after rounding.
- All core business analysis uses the 2021 to 2025 curated view rather than the complete public table.

SQL approach

The analysis uses beginner-friendly BigQuery SQL: SELECT, FROM, WHERE, SUM, COUNT, COUNTIF, DISTINCT, GROUP BY, ORDER BY, LIMIT, EXTRACT, ROUND, and simple comparisons. Complex window functions and nested analytical logic were intentionally avoided so every query remains explainable and reproducible.

4 Key findings

Annual market performance

Year	Net sales	Bottles	Million litres	Stores	Products
2021	\$428.12M	31.20M	24.76M	1,948	4,472
2022	\$428.78M	30.43M	23.57M	2,036	4,786
2023	\$446.07M	31.26M	23.73M	2,110	5,231
2024	\$447.24M	31.39M	23.31M	2,162	5,211
2025	\$424.99M	30.44M	22.19M	2,243	5,048

Sales peaked at \$447.24 million in 2024 and declined by \$22.25 million, or approximately 5.0%, in 2025. Active stores increased every year and were about 15% higher in 2025 than in 2021, yet 2025 sales finished slightly below 2021. Volume fell from 24.76 million litres in 2021 to 22.19 million litres in 2025, suggesting that price and product mix partly offset weaker physical volume.

Monthly pattern behind the 2025 decline

Month	2024	2025	Change
February	\$35.10M	\$31.31M	-\$3.79M
May	\$39.36M	\$35.59M	-\$3.77M
November	\$40.09M	\$34.75M	-\$5.34M
September	\$34.41M	\$35.16M	+\$0.75M
December	\$41.20M	\$41.75M	+\$0.55M

Monthly sales were calculated for all five years. The interpretation focused on 2024 and 2025 because 2024 was the peak year and 2025 recorded the subsequent decline. Sales were lower in 10 of 12 months in 2025, showing that weakness was broad rather than caused by one isolated season.

Category and product performance

Leading category	Net sales	Bottles	Sales per bottle
American Vodkas	\$323.38M	31.78M	\$10.18
Canadian Whiskies	\$239.16M	14.85M	\$16.11
Straight Bourbon Whiskies	\$189.42M	8.48M	\$22.33
100% Agave Tequila	\$151.67M	5.23M	\$29.00
Whiskey Liqueur	\$130.21M	21.83M	\$5.96

American Vodkas was the largest category by sales and bottle volume. Canadian Whiskies ranked second overall and generated a moderate \$16.11 per bottle. Agave Tequila generated the highest value per bottle despite much lower volume, while Whiskey Liqueur was strongly volume-driven. Sales per bottle is a category mix indicator, not a standardized price or profit measure.

Category	2024	2025	Change
American Vodkas	\$67.06M	\$64.94M	-\$2.12M
Canadian Whiskies	\$49.83M	\$45.80M	-\$4.03M
Straight Bourbon Whiskies	\$40.28M	\$39.98M	-\$0.30M
100% Agave Tequila	\$34.62M	\$33.37M	-\$1.25M
Whiskey Liqueur	\$26.52M	\$27.02M	+\$0.50M

Four of the five leading categories declined in 2025. Canadian Whiskies recorded the largest reduction at approximately \$4.03 million. The five categories together declined by about \$7.20 million, meaning other categories also contributed to the total market decline.

Leading product SKUs

Item	Product SKU	Net sales	Bottles
38178	Tito's Handmade Vodka	\$56.15M	1.97M
38177	Tito's Handmade Vodka	\$46.42M	2.37M
11788	Black Velvet	\$41.07M	2.51M
38176	Tito's Handmade Vodka	\$30.90M	2.06M
1799	Captain Morgan Original	\$29.07M	1.05M

Three Tito's Handmade Vodka item numbers ranked within the top four. The repeated description represents separate SKUs, likely product-size or package variants, and should not be treated as a consolidated brand total without a separate brand-mapping process.

Geographic and retailer performance

County	Net sales	Bottles	Active stores
Polk	\$505.92M	35.20M	344
Linn	\$184.29M	13.70M	184
Scott	\$141.41M	11.00M	126
Johnson	\$124.19M	7.96M	99
Black Hawk	\$111.42M	8.62M	121

Polk County generated \$505.92 million, approximately 2.7 times Linn County. The top ten counties represented about 65% of statewide sales. These are total market contributions, not per-capita consumption or store-efficiency measures.

Store	Retailer location	County	Net sales
2633	Hy-Vee 3 BDI Des Moines	Polk	\$69.76M
4829	Central City 2	Polk	\$65.84M
5916	Another Round DeWitt	Clinton	\$28.20M
3773	Benz Distributing	Linn	\$24.53M
3814	Costco Wholesale 788	Dallas	\$19.74M

The two leading purchasing locations were in Polk County and each generated more than twice the third-ranked store. However, the top ten locations represented only about 13.5% of total sales, so activity remained distributed across a broad retailer network.

5 Recommendations

- Prioritize recovery analysis for Canadian Whiskies and American Vodkas. Canadian Whiskies had the largest 2025 decline among the five leading categories, while American Vodkas remains the largest revenue and volume category.
- Use differentiated category strategies. Protect assortment and availability for high-value Agave Tequila and Bourbon products while maintaining dependable coverage for high-volume Vodka and Whiskey Liqueur products.
- Review February, May, and November planning. These months recorded the largest year-over-year dollar declines and may warrant closer examination of assortment, retailer orders, or market conditions.
- Prioritize large markets such as Polk, Linn, and Scott while avoiding excessive dependence on a small number of counties or retailer locations.
- Monitor top SKUs by item number. Multiple Tito's SKUs lead the ranking, so product-level availability matters; brand conclusions should be made only after product variants are mapped consistently.

6 Limitations

- The data represents wholesale liquor purchases by licensed Iowa retailers, not individual consumer transactions.
- The dataset does not include profit, operating cost, inventory, promotion, customer demographics, or competitor information, so the analysis cannot prove causes or profitability.
- County totals are not adjusted for population; they should not be interpreted as per-capita consumption.
- Sales per bottle is affected by product and bottle-size mix and is not a standardized shelf-price or margin measure.
- Negative paired sales and quantity values were retained as likely returns or corrections; the source does not provide a separate transaction-type field in the analysis view.
- The BigQuery Sandbox view is temporary and should be recreated from the saved SQL if it expires.

Conclusion

The project demonstrates a complete SQL workflow: source assessment, data-quality validation, construction of a reusable analysis view, focused business analysis, and evidence-based recommendations. The results show a stable market through 2024 followed by a broad 2025 decline, with clear differences across categories, products, counties, and retailer locations. The analysis remains intentionally concise and reproducible so each conclusion can be explained directly from the SQL outputs.

7 Dataset source and permitted use

Dataset: Iowa Liquor Retail Sales. BigQuery provider: Iowa Department of Commerce. Original data source: State of Iowa data portal. The BigQuery listing identifies the dataset as licensed under CC0 1.0 Universal, with the data provided as is. Attribution is included as a professional data-governance practice even though CC0 does not require attribution.

BigQuery table: bigquery-public-data.iowa_liquor_sales.sales

BigQuery Marketplace listing: [Iowa Liquor Retail Sales](#)

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